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Assessing the Associations Between Antiretroviral Treatment Resource Shortage and HIV Prevalence Among Teenagers residing in Zambia

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Abstract

Human Immunodeficiency Virus (HIV) is a global public health issue, with the highest burden in Sub-Saharan Africa. HIV is a virus that targets the immune system and specifically destroys CD4 T-cells. Over time, this puts the individual at risk of opportunistic infections. While Antiretroviral Therapy (ART) has been a significant factor in improving longer term survival rates and in reducing HIV transmission, the distribution of ART is not equally accessible in all countries. The literature has discussed several factors that contribute to the low rates of ART use, such as, social stigma, cost, and distance.

This paper examines if the lack of medical resources also plays a role in HIV prevalence among older adolescents (ages 15-19) in Zambia. Using data from the Zambia Demographic Health Surveys (2018-2024) and existing literature, this paper demonstrates that the lack of medical resources and the availability of ART are significantly correlated with lower access to treatment and higher rates of HIV prevalence among adolescents. Findings from this study underscore the importance of implementing improved access to health service, school-based HIV/AIDS education, and community-based ART delivery, would reduce the disparity in health outcomes and HIV burden.

I. Introduction

HIV (Human Immunodeficiency Virus) is a lifelong infection that affects the body's immune system. It can be spread through blood, semen, vaginal secretions, rectal fluids, and breast milk. Most commonly, the disease is spread through sexual intercourse. HIV can develop into AIDS (Acquired Immunodeficiency Syndrome) if left untreated, causing the immune system to become severely impacted.

Although there have been numerous efforts to prevent the transmission of HIV, adolescents are still a susceptible demographic for HIV infection. This is especially true in sub-Saharan Africa. Zambia continues to experience a significant HIV burden. Adolescents aged 15 to 19 represent a key at-risk group in Zambia. Many structural inequalities, such as economic deprivation, inadequacy of health facilities and health care access, and social stigmatization create inequalities in HIV prevention and care. Past studies have focused on the issue of stigma and the cost of health care as factors that prevent HIV care (Fox et al. 2010). However, the association between the lack of health care resources, including health care providers, antiretroviral therapy (ART), and health care facilities, and HIV outcomes have not been extensively examined.

The purpose of this study is to examine the association between lack of ART resources and HIV prevalence among adolescents aged 15-19 in Zambia. It is important to study this association to provide evidence that can guide public health policies to reduce the rate of HIV infections among adolescents and to increase the accessibility to treatment. As well as reduce the global health burden.

II. Problem statement

Despite ongoing efforts to reduce HIV transmission, adolescents aged 15–19 in Zambia continue to experience a significant HIV burden, and while past studies have focused on stigma, cost, and awareness as barriers to ART access (Fox et al., 2010), the association between lack of health care resources which include lack of health care providers, ART, health care facilities, and HIV outcomes has not been extensively examined. This lack of understanding makes it difficult to explain disparities in HIV prevalence and treatment access among adolescents in Zambia.

III. Literature review/evidence

Some studies have been conducted to understand the challenges people face in accessing ART in Zambia. Fox et al. (2010) conducted a cross-sectional analysis of ART and concluded that costs, stigma, and awareness were the major reasons for delayed initiation of ART in urban and rural areas of Zambia. In another study, Mwamba et al. (2022) concluded that people newly diagnosed with HIV usually wait to start treatment because of preparedness, fear, and health system barriers.

Dube et al. (2010) found that providing mobile ART services in rural areas of Zambia helped in overcoming the challenges associated with the lack of health facilities. Nichols et al. (2021) also found that the community-based ART distribution was cost-effective and provided better health outcomes.

Katirayi et al. (2021) concluded that fear of stigma and myths surrounding HIV discouraged the youth from getting tested and starting treatment. Miti et al. (2020) studied the

drug resistance among adolescents on ART and concluded that the drug resistance was a major concern in the adolescent population.

Although these studies explain the reasons for delayed initiation and adherence to ART, they fail to address the impact of the lack of medical resources (doctors, hospital beds, etc.) on the HIV-positive older adolescent population.

IV. Methods

This study will employ a secondary data analysis of the trend data from the Zambia Demographic Health Surveys, 2018 to 2024, as well as literature from peer-reviewed articles. The sample of interest are adolescents ages 15 to 19. The variables of interest are: Antiretroviral therapy (ART) availability, healthcare facility availability, and HIV prevalence. A qualitative systematic review will be performed to assess the relationship between health resource shortages and HIV-related outcomes.

V. Findings

There is a positive relationship between the lack of health resources and HIV prevalence among adolescents in Zambia. Regions with fewer facilities and less access to ART have lower proportions of treated HIV and higher proportions of untreated HIV.

Shortages in resources have a multiplicative effect and compound to affect access to treatment. Insufficient health facilities increases distance to care, which increases costs to care and decreases the likelihood an adolescent will seek treatment. Furthermore, a lack of healthcare

workers decreases the frequency and quality of care an adolescent may receive.

According to Dube et al. (2010), when barriers to care are reduced (e.g., utilizing mobile health clinics), proportions of adolescents on ART increase. Additionally, Nichols et al. (2021) found that decentralized community-based care resulted in better adherence and overall health outcomes. This suggests that the availability of resources plays a significant role in the likelihood an adolescent will receive HIV treatment.

VI. Discussion

While the literature is rich with evidence on the impact of structural barriers on HIV, this particular analysis contributes to the discussion on how shortages in resources is a key determinant of HIV that interacts with other barriers such as stigma and cost.

For adolescents, in particular, the lack of access to youth-friendly facilities combined with the fear of stigma results in a vicious cycle of fear to test for HIV and access treatment, ultimately leading to increased transmission and poor HIV-related outcomes among adolescents.

Thus, addressing shortages in resources may have a ripple effect of not only enhancing access to ART, but also mitigating the effect of stigma through increased ART uptake and normalization within the community. Enhancing the capacity of health facilities and ensuring a constant supply of ART are thus essential in reducing HIV infection. Specifically, the following interventions may be employed to overcome the challenges highlighted in this study:

VII. Proposed Interventions

1. **Increasing mobile ART clinics.** Enhancing the number of mobile clinics providing ART may enhance ART access among adolescents (Dube et al., 2010).
2. **Intensifying school-based HIV/AIDS education.** Knowledge and information on HIV/AIDS provided through education in schools can help reduce HIV/AIDS-related stigma and enhance HIV testing among adolescents (Katirayi et al., 2021).
3. **Implementation of community-based ART.** Provision of ART from community-based facilities has the potential of enhancing ART access and adherence (Nichols et al., 2021).
4. **Increasing funding for health facility infrastructure.** Funding for the construction of clinics, hiring of health care staff, and the supply of ART may help reduce shortages in resources and infrastructure. These strategies may be used in combination to achieve better results.

VIII. Conclusion

HIV/AIDS is a critical public health problem among adolescents in Zambia, and lack of resources is one of the key underlying factors to HIV/AIDS-related outcomes. The present study has established a significant relationship between shortage in ART and health facility infrastructure and HIV infection among adolescents. Consequently, a significant reduction in HIV infection among adolescents can be achieved through enhanced access to ART and health facility infrastructure via mobile clinics, community-based ART distribution strategies, and school-based HIV/AIDS education. Generally, enhancing the capacity of the health system remains a critical strategy toward global HIV/AIDS control.

IX. References

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